

Analyses: Classification with KNN

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Description

k-nearest neighbor analyses of the feature vectors per character sequence (loaded from ~/results/features_combined.csv). The results are stored in ~/results/classifier/.

Load Packages

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
library(class)
library(gmodels)
library(caret)
```

Load Data

Load data table with estimated feature values per sequence.

```
# load feature value estimations
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.8962406   1.5295563  0.41666667
## 2 Aurignacien      vvvvv         5  0.0000000   0.5711903  0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047  0.22222222
## 4 Aurignacien vvvvvvvvvvvvvv     14  0.0000000   0.6379550  0.07142857
## 5 Aurignacien      |||           3  0.0000000   0.4308271  0.33333333
## 6 Aurignacien      |||           4  0.0000000   0.5070802  0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Select subcorpora for binary classification.

```
# choose subcorpora to compare
subcorpus1 <- "TeDDi"
subcorpus2 <- "Cuneiform (Uruk V)"
selected <- c(subcorpus1, subcorpus2)
estimations.df <- estimations.df[estimations.df$subcorpus %in% selected, ]
```

Select relevant columns of the data frame, i.e. the measures to be included in classification and the “subcorpus” column.

```
estimations.subset <- estimations.df[c("subcorpus",
                                       "huni.chars",
                                       "hrate.chars",
                                       "ttr.chars",
                                       "rm.chars"
                                       )]
```

Remove NAs (whole row)

```
estimations.subset <- na.omit(estimations.subset)
```

Center and scale the data

```
# first column is subcorpus column, the others are the features
estimations.scaled <- cbind(estimations.subset[1], scale(estimations.subset[2:ncol(estimations.subset)]))
nrow(estimations.scaled)
```

```
## [1] 932
```

Create Training and Test Sets

```
# Generating seed
set.seed(1234)
# Randomly generating our training and test samples with a respective ratio of 2/3 and 1/3
datasample <- sample(2, nrow(estimations.scaled), replace = TRUE, prob = c(0.67, 0.33))
# Generate training set
estimations.training <- estimations.scaled[datasample == 1, 2:ncol(estimations.scaled)]
nrow(estimations.training)
```

```
## [1] 630
```

```
# Generate test set
estimations.test <- estimations.scaled[datasample == 2, 2:ncol(estimations.scaled)]
nrow(estimations.test)
```

```
## [1] 302
```

Get training and test labels

```
# Generate training labels
training.labels <- estimations.scaled[datasample == 1, 1]
```

```
# Generate test labels
test.labels <- estimations.scaled[datasample == 2, 1]
```

Initialize data frame

```
knn.results <- data.frame(k = numeric(0), accuracy = numeric(0), precision = numeric(0),
                          recall = numeric(0), f1 = numeric(0))
```

Building knn classifier

```
# choose maximum number of neighbors n
n = 10
# run a loop over different numbers of neighbors up to n
for (k in 1:n){
  # knn estimation of labels
  predictions.knn <- knn(train = as.data.frame(estimations.training),
                        test = as.data.frame(estimations.test),
                        cl = training.labels, k = k)

  # model evaluation
  # creating a dataframe from known (true) test labels
  test.labels <- data.frame(test.labels)
  # combining predicted and known classes
  class.comparison <- data.frame(predictions.knn, test.labels)
  # giving appropriate column names
  names(class.comparison) <- c("predicted", "observed")
  # inspecting our results table
  head(class.comparison)
  # get confusion matrix
  cm <- confusionMatrix(data = class.comparison$predicted,
                       reference = as.factor(class.comparison$observed))

  print(cm)
  # get precision, recall, and f1 from the output list of confusionMatrix()
  accuracy <- cm$overall['Accuracy']
  f1 <- cm[["byClass"]][["F1"]]
  recall <- cm[["byClass"]][["Recall"]]
  precision <- cm[["byClass"]][["Precision"]]

  # prepare data frame with results
  local.results <- data.frame(k, accuracy, precision, recall, f1, row.names = NULL)
  local.results.rounded <- round(local.results, 2)
  # print(local.results.rounded)
  knn.results <- rbind(knn.results, local.results.rounded)
}
```

```
## Confusion Matrix and Statistics
##
##               Reference
## Prediction   Cuneiform (Uruk V) TeDDi
```

```

## Cuneiform (Uruk V)          25    0
## TeDDi                      0   277
##
##           Accuracy : 1
##           95% CI : (0.9879, 1)
##       No Information Rate : 0.9172
##       P-Value [Acc > NIR] : 4.643e-12
##
##           Kappa : 1
##
## Mcnemar's Test P-Value : NA
##
##           Sensitivity : 1.00000
##           Specificity : 1.00000
##       Pos Pred Value : 1.00000
##       Neg Pred Value : 1.00000
##           Prevalence : 0.08278
##       Detection Rate : 0.08278
##   Detection Prevalence : 0.08278
##       Balanced Accuracy : 1.00000
##
##       'Positive' Class : Cuneiform (Uruk V)
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##
print(knn.results)

##      k accuracy precision recall f1
## 1    1          1          1      1  1

```



```
## 2 2 1 1 1 1
## 3 3 1 1 1 1
## 4 4 1 1 1 1
## 5 5 1 1 1 1
## 6 6 1 1 1 1
## 7 7 1 1 1 1
## 8 8 1 1 1 1
## 9 9 1 1 1 1
## 10 10 1 1 1 1
```

Write to file. Note that the file names have to be changed manually here.

```
write.csv(knn.results, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/results_KNN",
                                     subcorpus1, "_",
                                     subcorpus2, "_",
                                     ".csv"
                                     ), collapse = ""),
          row.names = F)
```